Activation generation We generated the activations for the required layers by extracting the activations from the last token in the residual stream, the same as in Section 2.1. However, here we use Contrastive Activation Addition (Rimsky et al., 2023). To get these steering vectors, we ran the model on the prompt matching the behaviour and on the prompt *not* matching the behaviour. The prompts were created by appending the corresponding answer to the template above. We got the activations for both prompts, and subtracted the non-matching activations from the matching activations for the 500 samples in the training split. This contrastive approach has been found effective in steering behaviours (Rimsky et al., 2023; Zou et al., 2023).

Activation injection For the individual steering vectors, we did a hyperparameter grid search to find the injection coefficients and the layer to inject in. The grid search was done both for adding and subtracting the steering vector during inference. The goal of this hyperparameter sweep is to find the largest difference between the default matching score and the score as a result of steering. The matching score is the share of answers that the (steered) model gives that match the target behaviour. We consider two extra criteria, which concern the model's functionality in answering the questions. First, if more than 5% of the output tokens do not match the possible options, the corresponding hyperparameters are discarded. Notably, we have never observed faulty answers with unsteered models. Second, steering too strongly can lead to cases where the model nearly always gives the same answer for each sample. Since the matching answers are distributed evenly between the two answer options, a heavily skewed answer distribution does not show a certain behavioural preference, but an incapable model. To avoid this mode collapse, the hyperparameter combination is discarded if one valid output token's frequency is >95% (e.g. A occurs 3 times and B occurs 197 times).

The steering vectors were combined into one steering vector in 8 different combinations, resulting from all the possible combinations of 3 binary differences. As a start, we either multiply each steering vector by their respective injection coefficient found by the grid search, or we multiply them by 1 (the injection coefficient varies between adding and subtracting, see Table 1). Secondly, we either take the mean of these activations, or we sum them up. Thirdly, we subtract or add the combined steering vectors, for a total of 8 combined steering vectors.

Furthermore, we also steer simultaneously at different places in the model, each regulated by the same global injection coefficient. We inject the myopia steering vector in layer 11, wealth seeking in layer 12, sycophancy in 13, agreeableness in 14, and anti-immigration in 15. We do this for global injection coefficients ranging from -2 to 2 with steps of 0.05.

Evaluation We used the matching score as metric. Too strong steering leads to faulty answers (answers not being yes/no or A/B) or mode collapse, as previously explained. For combined steering, if the model outputs too many faulty answers a score of -0.1 was given and otherwise if steering resulted in mode collapse a score of -0.2 was given.

For simultaneous steering, we calculate the matching scores for each behaviour while steering with the same vector. To measure the alignment tax, we also calculate the top-1 accuracy on 500k tokens from the Pile, containing both text and code. We calculate this top-1 accuracy while the model is being steered for each global injection coefficient.

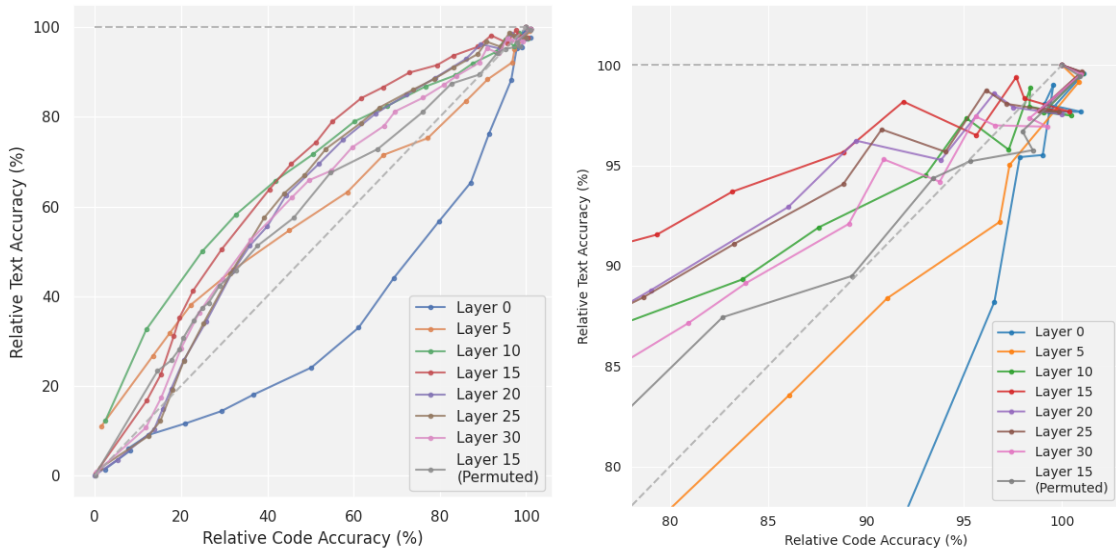
3 Results

3.1 Broad steering

Figure 3 shows the relative coding vs textual performance for top-1 next token prediction accuracy, and Figure 4 shows this for Python-specific performance. We first describe the results in Figure 3, and afterwards compare it with Figure 4.

After subtracting the coding steering vector and adding the text steering vector for various injection coefficients, we see that activation steering works to relatively reduce coding ability for most layers. For example, a 60% relative top-1 coding accuracy (40% fewer correctly predicted tokens) corresponds to an 80% relative textual accuracy for layer 15. Because developers will likely only accept a marginal penalty on general performance, Figure 3b, sheds light specifically on these smaller margins. We see a 10% reduction in coding performance, corresponding to only a 3% reduction in textual performance. The permuted steering vectors show a pattern similar to worse performing layers. The steering vector for layer 0 surprisingly produced the opposite of the intended effect.

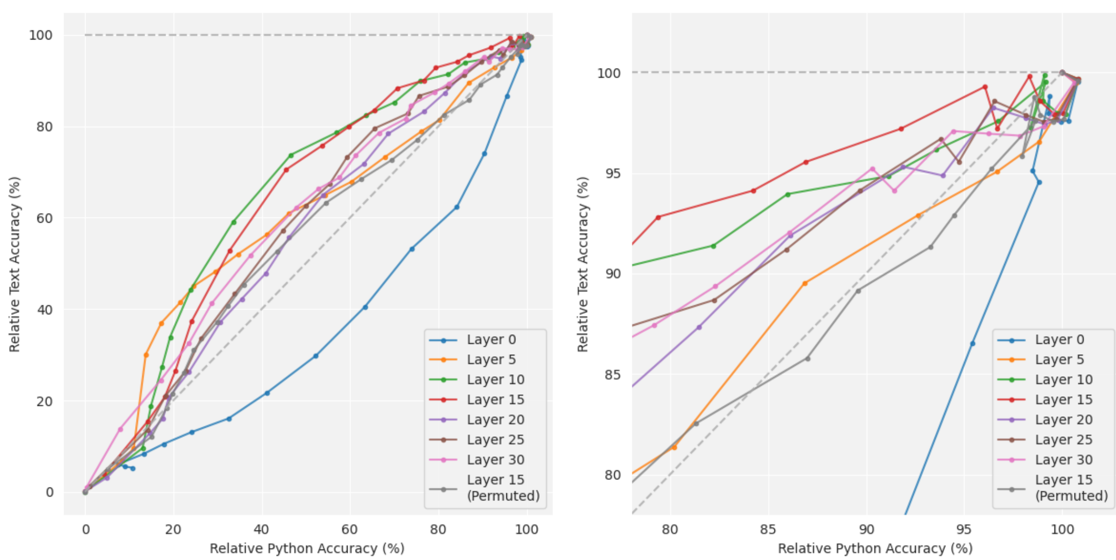
We hypothesized that steering for broader concepts would work but with a smaller effect size compared to narrower steering. Figure 4a shows similar results to Figure 3a, contradicting our hypothesis. In the cropped Figure 4b, we again see a similar performance to general coding. Although the steering vectors clearly work, the selective pruning method by Pochinkov and Schoots (2023) is substantially more effective.



(a) The overall performance.

(b) The same as in (a), but cropped to the top scores.

Figure 3: This figure illustrates the effect of applying a steering vector aiming to remove coding ability. The scores for text and code data are recorded. Each line represents steering at a certain layer, and each dot represents the scores for one injection coefficient. The horizontal dotted line illustrates perfect performance (which is impossible for our data due to noise), and the diagonal dotted line shows equal performance drops in code and text.



(a) Subfigure Python ability

(b) Python ability cropped.

Figure 4: This figure illustrates the effect of applying a steering vector aiming to remove Python ability, in the same way as in Figure 3.

3.2 Multi-steering

Here we show the results for layer 15, see Appendix B for the results for layer 10. In Figure 5 we see that for myopia, wealth seeking, agreeableness, and anti-immigration individual steering works in both directions but with varying effect sizes. Interestingly, steering for sycophancy has a negligible effect in our experiment, in contrast with previous work (Rimsky et al., 2023). The difference is likely due to our focus on political topology, and not other forms of sycophancy.

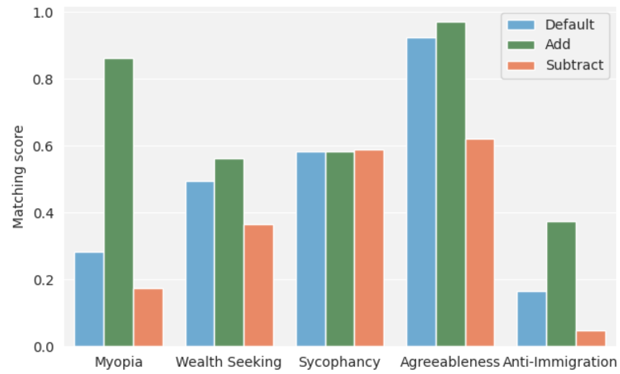


Figure 5: The results for individual steering for layer 15. The used injection coefficients can be found in Table 1

The results of the 8 combined steering vectors are shown in Figure 6. We observe that combined steering leads to unexpected and often smaller effect sizes than steering individually. The smaller effect sizes are clearly visible for the first three combinations for addition and subtraction. The last combination for myopia demonstrates unexpected effects: the weighted summation steering vector for addition has a larger effect size than with individual steering, but the subtracting weighted summation for subtraction leads to a higher matching score than without steering at all. Moreover, for anti-immigration, we see that adding generally leads to lower matching scores and subtracting to higher matching scores, which is also in contrast to expectation. For wealth seeking in particular, none of the combined steering vectors maintained a substantial effect. We see that combined steering vector leads to mode collapse in some cases, indicating an increased alignment tax.

Figure 6: Combining the individual steering vectors into one injected in layer 15. The combinations differ in three dimensions: take the mean or sum, weighted or unweighted, and subtracted or added. We compare the combined steering to the individual steering presented in Figure 5, which are indicated with the grey horizontal lines.

In Figure 7 we show the effect of simultaneous steering at multiple layers of the residual stream at the last token. For myopia and wealth seeking behaviours, the effects are comparable to individual steering. For anti-immigration and agreeableness the effect sizes are smaller and more unstable than with individual steering. For anti-immigration the steering only works for small injection coefficients, and for agreeableness the opposite effect occurs for small injection coefficients. As expected based on individual steering, sycophancy steering does not work here either. Additionally, as the absolute global injection coefficients increase, we see the effect of mode collapse arising. The scores converge to the dotted brown horizontal line at 0.5, which is the score when each answer has the same token, e.g. ‘Yes’. The alignment tax appears to be minor; there is only a couple percentage points decrease for global injections coefficients of -1 and +1.